

Talk Like a Structured Graph? Structural Statistics Help Small LLMs Mainly by Stating the Answer

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Abstract

Language models are increasingly used to reason over structured data, yet a graph written out as text is still hard for them to read, and their accuracy depends on how it is written. Do the structural encodings that help graph neural networks help here too? We state degree, clustering and random-walk returns as a primer before GraphQA’s incident encoding, and test Qwen3-1.7B and Qwen3-4B, thinking off and on, on paired graphs. Stated degrees give the clearest effects, in both directions: they raise node-degree accuracy where a model’s own count is unreliable and lower it for the model that already counts reliably. Clustering and RWSE state no answer; their effects are mostly small, change sign with task and model, and never significantly improve reading the queried node’s edges. A near-constant component-count sentence does for plain Qwen3-4B, which then restates that node’s line far more often. Checked against the graph, correct cycle answers can rest on nonexistent edges. Structural statistics help small models mainly by stating the answer, not by improving how they compute it: a primer is not a general aid, and whether it helps depends on whether the statistic states what the task asks and on whether the model can already compute it.¹

1 Introduction

Degree and random-walk encodings help graph neural networks and graph Transformers by supplying structure that message passing alone does not compute (Dwivedi et al., 2022; Rampásek et al., 2022). Language models are now applied to graph problems too: a frozen model needs no graph-specific training, takes any graph that can be written down, and answers a question posed in words. Written out, though, a graph is a list of edges, and the model must recover every structural quantity itself, with accuracy sensitive to the encoding (Fatemi

et al., 2024; Wang et al., 2023). The statistics that graph models receive are cheap to precompute, so the obvious aid is to write them into the prompt. We ask: *when a frozen language model reads a textual graph, do explicit structural statistics improve execution of graph queries?* Here execution means deriving an answer from edges rather than copying it from a primer (§3).

A *primer* is a preamble of plain sentences stating a structural statistic, usually one sentence per node, placed before GraphQA’s encoding and question (Figure 1); it changes what the model reads first, never the graph or the task. A primer can supply an answer, offer a clue, or alter how graph text is used. In an independent-edge graph every edge is independent of the others, so clustering and return probabilities carry little information about GraphQA’s answers beyond what they reveal of degrees and density (§4.1); a gain from them mostly reflects how the model reads the prompt. We expected a small model’s errors to come from computing local structure, not from lacking it, so that a stated statistic would raise accuracy on the tasks that need it, beyond what any added text does, and most where the model’s own count is least reliable. We test each primer against a graph-blind solver, a structure-free *filler*, and the same model’s thinking mode, which for Qwen3-1.7B uses 6.2–17.1 times as many output tokens on node degree. Degree, which supplies the queried value and the terms for edge count, is the positive control.

Contributions. (i) Stated degrees substitute for execution rather than improve it: they raise node-degree accuracy where a model’s own count is unreliable and lower plain Qwen3-4B’s where its count is reliable, and that loss runs through a change of procedure, from listing the node’s neighbors to reading a value off (§4.2). (ii) Clustering and RWSE, which state no answer, have mostly small effects whose sign depends on the model, and

¹<https://github.com/GalBrk/GraphTalk>

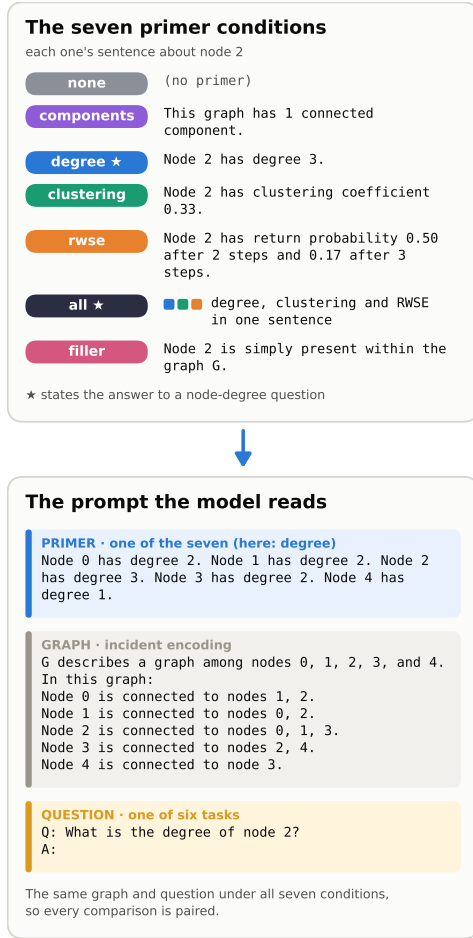


Figure 1: How a primer is used. Top: the seven conditions, each with its sentence about node 2 of a five-node example (the study’s graphs have 40 nodes); ★ marks the two that state a node-degree answer. Bottom: the prompt the model reads, one primer followed by GraphQA’s incident encoding and the question.

never significantly improve reading the queried node’s edges (§4.3, §4.6). (iii) A near-constant component-count sentence makes plain Qwen3-4B restate the queried node’s line and matches thinking on one task, while filler moves the same procedure the other way: the effect lies in how the text is read, not in what it says (§4.3, §4.4). (iv) Responses show stated values being used, at times copied from the wrong line, but not other statistics, and a correct cycle answer can rest on an edge the graph lacks (§4.6).

2 Related Work

Graphs as text. GraphQA (Fatemi et al., 2024) and NLGraph (Wang et al., 2023) show that accuracy depends on encoding, task and graph structure; NLGraph’s build-a-graph prompts also change the procedure by instruction. Both vary how the graph

or the instruction is written, so a gain cannot be traced to one structural quantity. We fix GraphQA’s encoding and tasks and vary only a preamble of named statistics.

Structure for LLMs. Structure can reach a model as learned soft tokens (Perozzi et al., 2024) or as motif descriptions in the prompt (Das et al., 2024), but structure written into a prompt tends to be read as context rather than as topology (Huang et al., 2024). None separates a statistic that states the answer from one that only describes the graph. A graph-blind solver that scores each primer’s text alone makes that separation for us, with degree as the positive control.

Controls and responses. Irrelevant text, input length and formatting change answers (Shi et al., 2023; Levy et al., 2024; Sclar et al., 2024), partial-input baselines expose answers an input gives away (Feng et al., 2019), and explanations need not reflect the computation behind them (Turpin et al., 2023). Accuracy alone therefore cannot say whether a primer helped a model read the graph, so we compare every primer against a structure-free filler and check every graph claim a response makes against the graph.

3 Method

Models and graphs. Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with thinking off and on, make four *arms* (1.7P, 1.7T, 4P, 4T in tables). Decoding is zero-shot and greedy, one response per prompt (budgets in Appendix A). A pilot on the published GraphQA graphs with Gemma 4 (Gemma Team, 2026) and Qwen3-8B/14B left most arms near ceiling (Appendix C), and even at 40 nodes Qwen3-8B answers nearly every node-degree query correctly (96.5–99.8%). We therefore use harder graphs and smaller models, so that accuracy without a primer leaves room to measure a change in either direction. Our independent-edge $G(n, p)$ graphs (Gilbert, 1959) have $n = 40$ at four edge densities up to $p = .50$ (the *main sweep*) and, for node degree and edge existence, three more up to $p = .85$. Every condition and arm sees the same 100 graphs per task and density, so every comparison is paired.

Tasks. GraphQA’s six tasks in its incident encoding (Figure 1). Four have answers that vary: node degree, connected nodes (the queried node’s neighbors, exact set match), edge count and edge

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	60.50	+7.5 ^f	+3.0	+4.0	+10.2 ^f	+1.8	-0.5
	1.7T	76.25	+8.8 ^f	+1.5 ^f	+2.5 ^f	+12.2 ^f	-1.0	-3.0
	4P	99.25	-6.5 ^f	-0.5	-2.2	-8.0 ^f	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2 ^f	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5 ^f	+1.2 ^f	+1.0 ^f	+2.0 ^f	+0.8 ^f	-6.2
	1.7T	78.75	+6.2	+2.0	+0.2	+6.2	+2.5	+2.5
	4P	86.00	-3.0 ^f	+1.0 ^f	+0.0 ^f	-2.8 ^f	+9.2 ^f	-12.5
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P	1.25	+3.0 [†]	-0.5 [†]	-0.8 [†]	+0.8 [†]	-1.0 [†]	-0.5 [†]
	1.7T	15.75	-3.2 [†]	-1.2 [†]	-3.2 [†]	+5.0 [†]	-0.8 [†]	-1.5 [†]
	4P	2.00	+27.2 ^f	-1.2 ^f	-1.5 ^f	+3.8	-1.2 ^f	+1.2
	4T	20.00	+8.5 [†]	+24.8 [†]	+10.2 [†]	+1.8 [†]	+24.8 [†]	+12.2 [†]
Edge exists	1.7P	70.50	+4.0 ^f	+3.8 ^f	+5.0 ^f	+11.5 ^f	-8.2 ^f	-15.8
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8	+4.0 ^f	-10.0	-10.0	-2.8 ^f	-7.0
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0

Table 1: Primary result: correct share without a primer (% of all prompts) and paired change under each primer (points), on the same 400 graphs per row ($p \leq .50$). Shaded: the primer supplies the degree answer or every term needed to compute edge count. Bold: $q < .05$ against no primer (BH within arm, task and control); ^f: $q < .05$ against filler; [†]: at least 15% truncation on either side, and no significance marks. Intervals and constant-answer tasks: Table 5; truncation: Table 8; balanced accuracy: Table 6. P/T: plain/thinking.

existence. Node count (always 40) and cycle check (always yes) have constant answers here (Appendix B); cycle check serves only to ask what correct answers rest on.

Primers and controls. Before the encoding we place no primer, one sentence per node giving its degree, local clustering coefficient or return probabilities after 2 and 3 steps (a truncated RWSE; Dwivedi et al., 2022), or *all* three (Appendix A). Two diagnostics are a short sentence giving the number of connected components, and a filler that names every node but gives no graph statistic. The filler is about as long as clustering, longer than degree and shorter than RWSE and all three (Appendix Table 4). The graph-blind solver answers from primer text alone.

Outcomes and statistics. Each response is *correct*, *wrong* or *truncated* (it used its whole budget); shares use all responses, and an effect is the paired change in the correct share, in points. Cells with at least 15% truncation on either side are flagged ([†]) and left out of pooled summaries. Edge existence is also scored by balanced accuracy, because the share of queried pairs that are edges tracks the edge density (10% to 85%). The primary analysis is Table 1: each primer against no primer and against filler on the same 400 main-sweep graphs. It uses exact McNemar tests (McNemar, 1947), bootstrap intervals over graphs and Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) within each (arm, task, control) family; $q < .05$ is significant. Plain arms against

thinking (Table 2) use the same tests on the same graphs, and we call them equivalent when the 90% interval of their difference lies within ± 3 points, about the 4B arms’ median detectable gain. Analyses of mechanism are secondary and name their own test. With a median detectable gain of 3.2 to 5.7 points, a non-significant effect is unresolved, not zero (Appendix A).

Execution. Node-degree and neighbor-set queries ask about the same node. Our measure of execution is the *joint* share of nodes whose two answers are both correct: a stated degree can fix the first answer but not the second, while reading the node’s line correctly gives both.

Response readers. Rule-based readers record whether a response reads a stated value off (a *look-up*) or lists the neighbors, whether it reports a conflict with the primer, and which cycles and edges it asserts, checked against the graph. The conflict readers agree closely with blind hand labels (90–100% of 226 items); the others were spot-checked by an LLM under human supervision (Appendix A).

Follow-ups. Qwen3-1.7B node degree, plain and thinking, 400 graphs per density at all seven densities, a fresh seed, and a 20–160-node grid at mean degree 8 and 16 (also Qwen3-8B).

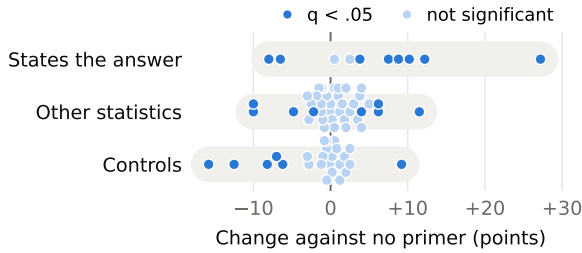


Figure 2: Every unflagged cell of Table 1, one dot per arm, task and primer, grouped by what the primer states. *States the answer*: degree and all three statistics on node degree and edge count; *Controls*: components and filler. Filled: $q < .05$ against no primer.

4 Results

4.1 What each primer states

The graph-blind solver recovers node degree and edge count perfectly from stated degrees, directly or by halving their sum. Otherwise it gains little over no primer: at most 21.0% on degree and 10.4% on edge count, against 14.4% and 3.0% (Appendix Table 3). These scores measure its rules, not every possible use of the text (Feng et al., 2019): in sparse graphs the rounded values partly reveal a node’s degree, and a lookup from printed value to degree, fit on other graphs, gets 54.0% from clustering at $p = .20$ and 41.0% from RWSE at $p = .10$, against 16.0% and 17.0% for the majority answer. Averaged over densities the lookup is far weaker, so neither statistic states the degree. Unshaded cells in Table 1 are *side effects*. Stated answers give the largest gains; other statistics and the controls move cells in both directions by similar amounts (Figure 2).

4.2 Stated answers are read off

The degree primer raises node-degree accuracy for plain and thinking 1.7B by 7.5 and 8.8 points and lowers it by 6.5 for plain 4B, which already counts correctly. Its gains are largest at intermediate unaided accuracy; at high unaided accuracy it can be a loss (Appendix Figure 3). On a log-odds scale, which does not compress near ceiling, the loss is plain 4B’s alone: at matched unaided accuracy, stated answers break more of its answers than they fix and fix more of plain 1.7B’s than they break. The same pattern holds within one model. The degree primer hurts plain 4B on the lowest-degree queried nodes in each graph and helps on the highest-degree ones, which have the most neighbors to count (−11.8 and +10.6 points).

Reading a value off is itself fallible. Across all seven densities, where the answer is stated, plain 1.7B gets it right less than half the time (43%). Plain 4B stops listing the node’s neighbors and reads the value off in most responses at $p = .50$ (62%). A third of its wrong look-ups copy the degree printed one sentence away, more often than chance predicts (Appendix B). The degree primer’s effect runs through this change of procedure: plain 4B’s loss comes almost wholly from graphs on which its route changed, and thinking 1.7B’s gain from graphs on which it did not (§4.5).

4.3 Side effects include gains and harms

Filler alone changes up to a fifth of the outcomes in an arm, often on questions primers also change. Identical-prompt rerun flips account for only a minority of each condition’s changes (Appendix B).

Side effects differ in sign. On plain 4B edge existence, clustering raises balanced accuracy and RWSE lowers it (+3.0 and −6.5 points); all three together raise plain 1.7B’s but lower plain 4B’s (all $q < .05$; Appendix Table 6). RWSE and all each cost plain 4B 10.0 points on edge existence. On its dense degree queries, degree alone gains 9.3, but all loses 21.3 on the same graphs (Appendix B). The largest unflagged gain in Table 1 from a primer without degrees, +9.2 on plain 4B neighbor sets, comes from the component sentence. All sampled graphs from $p = .20$ are connected, so the sentence is identical across them. With it the model almost always restates the queried node’s line, against about two thirds of responses with no primer and a fifth with filler (98.5%, 63.0%, 21.8%). Filler moves restatement the other way and costs the same cell 12.5 points.

The one side effect that replicates, clustering on plain 1.7B’s node degree, is small, specific to that model and to low density, and shows little sign of using the stated values (Appendix C).

4.4 Primers against the thinking mode

Where thinking leads (Table 2), the component sentence brings plain 4B’s neighbor-set exact match within the ± 3 -point equivalence margin, on a cell that filler also moves (§4.3). Elsewhere primers narrow the gap without closing it. All three statistics cut plain 1.7B’s node-degree gap to about a third (15.8 to 5.5 points). Under the degree primer its neighbor sets are not significantly below thinking, but the equivalence interval extends beyond the margin. Clustering and RWSE close less, and leave

Task	Model	% correct		Gap to thinking (points)		
		Plain	Think.	None	Best primer	Cl./RW.
Degree	1.7B	60.50	76.25	+15.8	all	+5.5
	4B	99.25	97.00	-2.2	components	-2.5
Neighbors	1.7B	72.50	78.75	+6.2	degree	+2.8
	4B	86.00	95.50	+9.5	components	+0.2
Edge count	1.7B	1.25	15.75 [†]	+14.5	degree	+11.5
	4B	2.00	20.00 [†]	+18.0	degree	-9.2
Edge exists	1.7B	70.50	99.00	+28.5	all	+17.0
	4B	90.50	99.75	+9.2	clustering	+5.2

Table 2: Can a primer close the plain mode’s gap to the same model thinking without a primer? *Plain, Think.*: correct share without a primer ($p \leq .50$, same graphs). *Gap*: thinking minus plain, in points, first with no primer (*None*), then under the plain model’s best primer and under the better of clustering and RWSE (*Cl./RW.*). A gap near 0 is closed; a negative gap puts the plain mode ahead. **Bold**: $q < .05$. Chosen on held-out graphs, the best primer is the same in every row. [†]: thinking truncates on most prompts (Table 8).

the plain mode significantly below thinking whenever thinking leads. The degree primer lifts plain 4B’s edge count above thinking. Thinking there is right on nearly every answer it finishes (97.6%), though, so this compares budgets, not ability, and the stated degrees already give the edge count.

4.5 Primers in the thinking mode

Near ceiling, thinking 4B moves by at most 2.5 points under any primer on node degree, edge existence and neighbor sets. Thinking 1.7B gains from stated degrees (Table 1) but checks them against its own count instead of reading them off. It keeps listing neighbors, and when its count conflicts with the stated value, it ends on the stated value in two thirds of cases. Checking costs budget: across all seven densities, its node-degree responses that run out of tokens rise from 1.1% to 7.6% under the degree primer. Where the budget binds, much of any primer’s effect is on finishing. Most of thinking 4B’s changed answers involve a response that ran out of tokens, and on its edge count even filler gains by lowering truncation (Table 8).

4.6 What correct answers rest on

Does a primer improve reading the graph itself? By the joint share (§3), clustering and RWSE never do significantly; the degree primer lowers plain 4B’s by 9.2 points and the component sentence raises it by the same amount (Appendix B).

Cycle check has the same “yes” answer on every sampled graph, but a response can justify it by a real cycle or by the edge count. Without a

primer, valid arguments back 34.2% and 58.5% of all prompts for the plain models, and most prompts with thinking. Plain 1.7B rests 43.5% of its correct answers on an invented cycle, one with a step that is not an edge (example in Appendix B). The degree primer raises its valid share to 50.2% yet lowers its accuracy by 17.5 points. The plain models switch to arguing from the number of edges (Table 7), a valid route that, like a look-up, need not read the edges. RWSE, all three statistics and filler lower plain 4B’s valid share.

5 Discussion

A stated degree is worth what counting it from the incident list costs (§4.2), and a look-up is not free either: it must match the right node to the right sentence, and wrong look-ups show where that fails.

Some structural encodings help message-passing GNNs by supplying what they cannot compute: return probabilities separate nodes that 1-WL-bounded networks cannot (Xu et al., 2019; Dwivedi et al., 2022). A language model has the information instead: every edge is in its context, and thinking 4B answers nearly all degree, neighbor-set and edge-existence queries (Table 2). The gap to thinking suggests that the plain models’ bottleneck is computation, not information, so a statistic can help by stating the answer or by changing how the text is read. Clustering and RWSE state no answer: their text reveals degrees mainly in sparse graphs (§4.1) and almost nothing about edge existence (Table 3). Their effects, which vary in sign, mostly reflect how the text is read.

6 Future Work

First, corrupt or omit the queried node’s stated degree, keeping all else fixed, to separate following a stated value, checking it and counting from edges. Second, shuffle node labels and primer order, and test the component sentence against neutral text of equal length and position; the same controls apply to clustering’s low-density degree gain. Third, sample acyclic as well as cyclic graphs, and families that vary triangles at matched size and density, so that cycle check has two answers and clustering carries information. Other model families, larger thinking budgets and a graph-counting tool would show when a primer is the cheapest way to supply a computation.

7 Conclusion

We asked three questions. *Does a structural primer help?* Mainly by stating the answer: a stated degree saves an unreliable count and disturbs a reliable one, and adding clustering and RWSE need not preserve its effect. *Does matching the statistic to the task matter?* Yes: clustering and RWSE, which state no answer, have mostly small effects of either sign and never significantly improve execution; plain Qwen3-1.7B’s rises by about 4 points under both, below what we can detect; the component sentence changes plain Qwen3-4B’s procedure instead. *Does capacity matter?* Through unaided accuracy: stated degrees help Qwen3-1.7B and hurt plain Qwen3-4B, yet also help plain Qwen3-4B on dense graphs, where its own count fails (Appendix B). Thinking mostly keeps its lead. Beside paired controls, completion rates and graph-checked claims show whether an answer was supplied, the incident line read, or a correct label validly argued.

Limitations

Scope: two Qwen3 models (and Qwen3-8B in one follow-up); independent-edge graphs of 40 nodes (20–160 in one follow-up); one encoding, one wording and position per primer; zero-shot; fixed token budgets. *Controls:* filler matches only clustering’s length; component count is structural information; no primer has shuffled or wrong values, and no instruction-only baseline isolates procedural wording. Nodes 0–9, listed first, also have the shortest ids, so position and id length are confounded. *Measurement:* response readers other than conflict detection were spot-checked by an LLM, not blind hand labels.

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A Method Details

Statistics. With adjacency matrix A and transition matrix $P = D^{-1}A$, the per-node statistics are $d_v = \sum_u A_{vu}$, $c_v = 2t_v/(d_v(d_v - 1))$ and $r_v^{(k)} = (P^k)_{vv}$ for $k \in \{2, 3\}$, where t_v counts edges among v 's neighbors and $c_v = 0$ when $d_v < 2$. Clustering and return probabilities are printed to two decimals, one sentence per node, in node order (Table 4).

Why these features. Degree and RWSE are the structural encodings graph Transformers use (Ram-pásek et al., 2022). Local clustering adds how a node's neighbors connect. The component count is a single global fact that, with the node and edge counts, decides whether a cycle exists. Laplacian encodings are excluded because eigenvector signs and bases need an explicit convention (Dwivedi et al., 2022). RWSE keeps only $k = 2, 3$: $r_v^{(1)} = 0$ in every simple graph, $k = 3$ detects triangles, and within a graph degree and clustering already explain most of the variance of $k = 4$ and $k = 5$.

Models, budgets and power. Models are the Qwen3 checkpoints on Hugging Face, run with transformers. Each response may use up to 8,192 new tokens, and 2,048 for the plain arms at $p \geq .65$, where no arm truncates more than 2.5% of its responses. The main sweep and its extension hold 84,000 responses on 700 graphs. In one 100-graph cell the median smallest detectable paired effect is 7.4 points, and 13.7 where side information meets intermediate accuracy. Under one BH correction over all 264 tests, every bold cell of Table 1 but one and all but three ^f marks stay significant. Greedy generation is not exactly reproducible: generating the same prompts twice changes the outcome of 5.3%, so we compare net paired changes, never counts of changed answers.

Graph-blind solver. It reads only the primer text, with 16 exact rules, 1 heuristic and 8 rules fitted on graphs disjoint from those scored (Table 3).

Response readers. A response is a look-up when it answers without restating the queried node's neighbor list, and lists the neighbors when it gives five or more one per line. The cycle and edge readers check every asserted step against the prompt's graph.

Scoring. The answer extractor parses 99.84% of responses. Set-F1 on neighbor sets is secondary

Primer	Degree	Neighbors	Edge count	Edge exists
none	14.4	0.0	3.0	73.5
degree	100.0	0.1	100.0	74.6
clustering	14.4	0.0	10.4	72.7
RWSE	21.0	0.1	3.0	55.4
all	100.0	0.8	100.0	74.1
components	14.4	0.0	3.0	73.5
filler	14.4	0.0	3.0	73.5

Table 3: Graph-blind solver accuracy (%) from the primer text alone, 40-node graphs, averaged over all seven densities ($p = .10-.85$), on the four tasks whose answer varies. RWSE falls below no primer on edge existence because a fitted density threshold fails at $p \geq .65$ (38.0, 48.0 and 12.0%).

Primer	Mean chars	Information
degree	891	Degree for each node
clustering	1,629	Local clustering for each node
RWSE	2,949	Two rounded return values per node
all	4,691	Degree, clustering and RWSE
components	37	Number of connected components
filler	1,829	Node names, no graph statistic

Table 4: Mean preamble length in characters over the main-sweep graphs.

because it gives partial credit: it moves in the same direction as exact match in 3 of 5 large contrasts, and by 0.11 to 0.17 as much. Balanced-accuracy changes use a paired permutation test with Benjamini–Hochberg adjustment over the six conditions within an arm.

B Detailed Results

Tables 5 and 8 complete Table 1 with 95% intervals, truncated shares and the two constant-answer tasks. The paragraphs below give the numbers Section 4 summarizes, in its order.

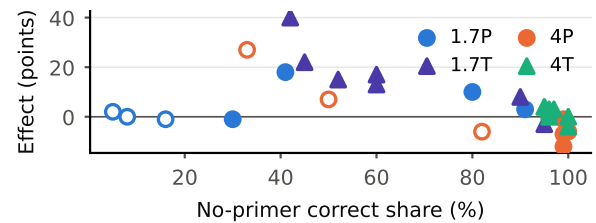


Figure 3: The degree primer's effect on node degree against the no-primer correct share, per arm and density. Hollow: plain arm at $p \geq .65$ (smaller budget).

Stated answers (§4.2). The gain follows the room the model leaves (Figure 3). In node-degree cells answered 25–50% and 50–75% correctly

Task	Arm	degree	clustering	RWSE	all	components	filler
Degree	1.7P	+7.5 [+3.0, +11.8]	+3.0 [-1.5, +7.5]	+4.0 [-0.5, +8.5]	+10.2 [+5.2, +15.2]	+1.8 [-2.2, +5.8]	-0.5 [-4.2, +3.5]
	1.7T	+8.8 [+4.2, +13.2]	+1.5 [-2.5, +5.5]	+2.5 [-2.0, +6.8]	+12.2 [+8.0, +16.5]	-1.0 [-4.2, +2.5]	-3.0 [-7.0, +0.8]
	4P	-6.5 [-9.2, -4.0]	-0.5 [-2.0, +0.8]	-2.2 [-4.2, -0.5]	-8.0 [-11.0, -5.2]	+0.2 [+0.0, +0.8]	-0.5 [-2.0, +1.0]
	4T	+2.5 [+0.8, +4.2]	+1.8 [-0.2, +3.8]	-1.2 [-3.8, +1.0]	+0.5 [-1.5, +2.8]	+0.2 [-1.8, +2.5]	+2.0 [+0.5, +3.8]
Neighbors	1.7P	+3.5 [-0.5, +7.2]	+1.2 [-3.0, +5.5]	+1.0 [-3.2, +5.2]	+2.0 [-2.5, +6.5]	+0.8 [-2.5, +4.5]	-6.2 [-10.5, -2.2]
	1.7T	+6.2 [+2.8, +10.0]	+2.0 [-2.0, +6.0]	+0.2 [-3.5, +4.0]	+6.2 [+2.5, +10.0]	+2.5 [-0.8, +6.0]	+2.5 [-1.0, +6.0]
	4P	-3.0 [-7.0, +1.0]	+1.0 [-2.8, +4.8]	+0.0 [-3.8, +4.0]	-2.8 [-7.0, +1.8]	+9.2 [+6.0, +12.8]	-12.5 [-17.2, -7.8]
	4T	+0.5 [-2.2, +3.2]	-0.8 [-3.2, +1.8]	+0.5 [-1.8, +3.0]	-1.2 [-3.8, +1.2]	+0.5 [-1.8, +2.8]	+1.2 [-1.0, +3.5]
Edge count	1.7P	+3.0 [†] [+0.8, +5.2]	-0.5 [†] [-2.0, +1.0]	-0.8 [†] [-2.0, +0.5]	+0.8 [†] [-1.0, +2.5]	-1.0 [†] [-2.2, +0.0]	-0.5 [†] [-1.8, +0.8]
	1.7T	-3.2 [†] [-7.8, +1.2]	-1.2 [†] [-5.5, +3.0]	-3.2 [†] [-7.2, +0.5]	+5.0 [†] [+0.2, +10.0]	-0.8 [†] [-5.0, +3.8]	-1.5 [†] [-5.5, +2.5]
	4P	+27.2 [+22.8, +32.0]	-1.2 [-3.0, +0.2]	-1.5 [-3.0, -0.2]	+3.8 [+1.2, +6.2]	-1.2 [-2.8, +0.0]	+1.2 [-0.8, +3.0]
	4T	+8.5 [†] [+3.2, +14.2]	+24.8 [†] [+19.5, +30.2]	+10.2 [†] [+5.5, +15.0]	+1.8 [†] [-3.5, +7.2]	+24.8 [†] [+19.8, +30.0]	+12.2 [†] [+7.2, +17.0]
Edge exists	1.7P	+4.0 [-0.2, +8.2]	+3.8 [-0.5, +8.0]	+5.0 [+0.2, +9.5]	+11.5 [+6.2, +16.5]	-8.2 [-11.8, -4.8]	-15.8 [-20.0, -11.8]
	1.7T	-2.5 [-4.5, -0.8]	-1.0 [-2.8, +0.5]	-1.2 [-3.0, +0.5]	-1.8 [-3.8, +0.0]	-0.8 [-2.5, +0.8]	-0.5 [-2.0, +1.0]
	4P	-4.8 [-7.8, -1.5]	+4.0 [+1.0, +7.0]	-10.0 [-13.8, -6.2]	-10.0 [-14.0, -6.0]	-2.8 [-5.8, +0.2]	-7.0 [-10.5, -3.5]
	4T	+0.0 [-0.8, +0.8]	-0.8 [-1.8, +0.0]	-1.2 [-2.5, +0.0]	-1.0 [-2.2, +0.0]	-0.5 [-1.5, +0.5]	+0.0 [+0.0, +0.0]
Node count	1.7P	+6.8 [+3.2, +10.5]	+67.2 [+64.5, +70.0]	+2.0 [-3.2, +6.8]	+47.0 [+42.5, +51.2]	+21.2 [+16.2, +26.2]	+39.8 [+36.0, +43.5]
	1.7T	+5.5 [+2.5, +8.5]	+9.0 [+6.5, +11.8]	+9.0 [+6.5, +11.8]	+9.0 [+6.5, +11.8]	+1.5 [-2.2, +5.0]	+9.0 [+6.5, +11.8]
	4P	+8.0 [+5.2, +10.8]	+8.8 [+6.0, +11.5]	+9.2 [+6.5, +12.0]	+9.8 [+7.2, +12.5]	+9.0 [+6.5, +11.8]	+9.2 [+6.8, +11.8]
	4T	+0.2 [-0.5, +1.2]	+0.2 [-0.5, +1.2]	+0.2 [-0.5, +1.2]	+0.5 [+0.0, +1.2]	+0.0 [-1.0, +1.0]	+0.5 [+0.0, +1.2]
Cycle	1.7P	-17.5 [†] [-23.0, -12.2]	+8.5 [+5.8, +11.2]	-0.2 [-4.2, +3.5]	+7.0 [+4.0, +10.2]	+2.0 [-1.5, +5.3]	+3.0 [-0.8, +6.8]
	1.7T	-21.8 [†] [-26.5, -17.5]	-0.5 [-2.8, +1.8]	+2.2 [+0.5, +4.0]	-0.8 [-3.2, +1.8]	-7.5 [-11.0, -4.2]	+3.0 [+1.5, +4.8]
	4P	-2.2 [-4.2, -0.5]	+0.5 [-0.5, +1.8]	-0.2 [-1.8, +1.2]	-0.2 [-1.8, +1.0]	-2.0 [-4.0, -0.2]	-0.5 [-2.0, +0.8]
	4T	-5.5 [-8.0, -3.0]	+0.0 [-1.2, +1.2]	+0.2 [-0.8, +1.5]	-0.5 [-2.0, +0.8]	-3.2 [-5.2, -1.2]	+0.5 [-0.5, +1.5]

Table 5: Every main-sweep effect against no primer (points) with its 95% bootstrap interval over graphs, 400 paired graphs per cell ($p \leq .50$). [†]: at least 15% truncation on either side. Node count (always 40) and cycle check (always yes) cannot measure reading the graph at 40 nodes. P/T: plain/thinking.

without them, answer-stating primers gain 16.1 and 12.9 points; at 75–90% they lose 5.8. These bands hold on held-out halves of the graphs and with the truncation flag at 10% or 50% (at least +12.5 in every case). An effect mostly follows how well the model does unaided. At a fixed accuracy without a primer, density adds little ($p = .28$ for answer-stating primers, $p = .85$ for side statistics), except that side statistics’ effect on node degree falls with density ($p = .006$). One model falls on both sides. The degree primer costs plain 4B 12.0 points at $p = .50$, where it answers 99.0% correctly unaided, and gains 27.0 at $p = .85$, where it answers 33.0% (both $q < .01$). Stated answers are not always taken: across all seven densities, where the solver scores 100%, node-degree accuracy under the degree primer is 43% and 81% for plain 1.7B and plain 4B, and 79% and 98% with thinking. At high density, where plain 1.7B is weakest, the stated degree barely helps it (at most 2 points). At $p = .85$ it still answers 39, the degree of a node joined to all others, on 70.8% of finished responses. Plain 4B’s wrong look-ups copy a nearby line: 17 of 57 equal the degree printed one sentence away, against 10.4 expected by chance ($p = .017$). Adding clustering and RWSE to the same degrees costs dense plain 4B the most (Figure 4): pooled over $p \geq .65$, degree alone gains 9.3 points ($q = .03$) and all three statistics lose 21.3 ($q < .001$).

Filler and fragile questions (§4.3). Filler changes 6.2–18.6% of outcomes per arm. A primer changes 38.5–69.1% of the questions filler changes

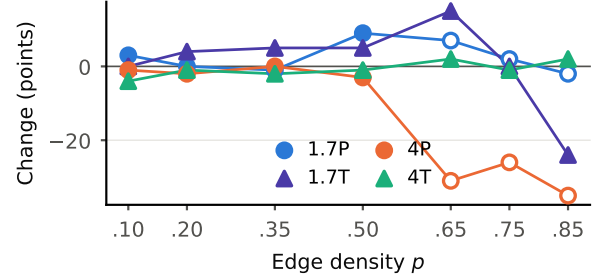


Figure 4: What adding clustering and RWSE to the same degrees changes on node degree (all three statistics minus the degree primer, points), per arm and density. Hollow as in Figure 3.

and 3.1–15.2% of the rest. Some questions are fragile: on plain 1.7B’s node degree, 24 of 400 change outcome when the same prompt is generated again. Every condition changes them far more often than the rest (45.8–66.7% against 14.4–26.1%, $q \leq .018$). Yet they carry only 11 to 16 of each condition’s 65 to 113 changes, a lower bound on chance with only two generations per prompt. Where accuracy without a primer is 25–75%, components, clustering and RWSE average +1.6 points while filler averages -6.9 on the same cells.

Side statistics (§4.3). On edge existence, every primer that moves balanced accuracy does so in a plain arm (bold in Table 6); clustering and RWSE on plain 4B have $q \leq .018$, and all raises plain 1.7B’s by 7.8 points and lowers plain 4B’s by 6.5. Plain 1.7B’s false alarms rise steeply with the number of neighbors a non-edge pair shares, from 1.7% with none to 65.9% with four or more, and plain

Arm	Primer	Raw	Balanced	Yes
1.7P	none	70.50	79.86	56.2
	degree	74.50	82.30	51.7
	clustering	74.25	82.42	52.5
	RWSE	75.50	82.68	50.1
	all	82.00	87.71	44.8
	components	62.25	74.23	64.5
1.7T	filler	54.75	69.11	72.0
	none	99.00	99.02	27.1
	degree	96.50	97.61	29.9
	clustering	98.00	98.63	28.6
	RWSE	97.75	98.17	28.3
	all	97.25	98.12	29.5
4P	components	98.25	98.81	28.3
	filler	98.50	98.68	27.8
	none	90.50	93.22	35.7
	degree	85.75	90.27	40.7
	clustering	94.50	96.25	32.3
	RWSE	80.50	86.69	46.3
4T	all	80.50	86.69	46.1
	components	87.75	91.64	39.0
	filler	83.50	88.74	43.2
	none	99.75	99.83	26.8
	degree	99.75	99.83	26.8
	clustering	99.00	99.32	27.0
	RWSE	98.50	98.68	26.9
	all	98.75	99.15	27.1
	components	99.25	99.49	27.0
	filler	99.75	99.83	26.8

Table 6: Edge existence, main sweep (400 prompts per row): raw and balanced accuracy over all prompts and the share of finished answers that say yes, all in %. Truncation is in Table 8. Bold: balanced accuracy differs from the same arm without a primer (paired permutation, BH $q < .05$ within arm).

4B’s from 0.2% to 38.8%. In $G(n, p)$ shared neighbors say nothing about whether a pair is joined, so this is consistent with mistaking nearby ids for adjacency. Under clustering, plain 1.7B’s responses explicitly claiming a non-edge is an edge fall from 13.7% to 4.4%, while its false alarms fall only from 40.3% to 35.2%: stated reasons move more than decisions. The component sentence raises plain 4B’s neighbor-set Set-F1 from 0.987 to 0.998, and responses that restate the queried node’s line are right 95%, 92% and 84% of the time with it, with no primer and with filler. At high density the printed statistics vary less within a graph: clustering values have SD 0.021 at $p = .65$, and rounded RWSE takes 3.30 distinct value pairs per graph at $p = .50$, against 12.50 distinct degrees. Clustering still gains 11.3 on dense plain 4B node degree with similar rates of neighbor enumeration, making presentation a plausible contributor.

Against the thinking mode (§4.4). Equivalence in Table 2 rests on 90% intervals. The component

Arm	Primer	Correct “yes” (n)	Cycle named			No cycle	
			Real	Invented	Not a cycle	All	Edge count
1.7P	none	363	36.9	43.5	16.0	3.6	0.8
	degree	293	13.0	20.5	1.4	65.2	55.6
	clustering	397	39.5	48.1	7.8	4.5	1.0
	RWSE	362	33.7	40.3	19.9	6.1	1.7
	all	391	38.4	50.4	7.9	3.3	1.3
	components	371	27.8	40.4	25.3	6.5	4.3
1.7T	filler	375	33.3	55.5	9.6	1.6	0.0
	none	388	79.9	17.8	1.0	1.3	0.5
	degree	301	61.5	29.6	0.0	9.0	9.0
	clustering	386	80.6	19.2	0.0	0.3	0.3
	RWSE	397	68.0	20.7	10.6	0.8	0.5
	all	385	73.0	21.8	3.1	2.1	1.8
4P	components	358	74.9	22.6	0.0	2.5	2.0
	filler	400	77.2	19.8	2.5	0.5	0.0
	none	396	34.1	23.0	6.8	36.1	25.0
	degree	387	13.4	3.9	2.1	80.6	65.6
	clustering	398	48.7	31.2	1.0	19.1	5.8
	RWSE	395	33.2	20.8	38.7	7.3	0.5
4T	all	395	45.1	33.4	6.6	14.9	0.8
	components	388	39.2	18.6	3.4	38.9	26.5
	filler	394	44.2	28.2	5.3	22.3	2.3
	none	397	96.5	3.3	0.0	0.3	0.0
	degree	375	90.9	5.1	0.0	4.0	2.7
	clustering	397	95.5	4.5	0.0	0.0	0.0
	RWSE	398	55.0	2.5	42.5	0.0	0.0
	all	395	92.7	6.3	0.0	1.0	1.0
	components	384	92.2	4.4	0.0	3.4	2.9
	filler	399	94.2	5.0	0.5	0.3	0.0

Table 7: What the correct cycle-check answers rest on (main sweep; every correct answer is “yes”). *Correct (n)*: count of correct, finished answers; the other columns are % of those n , and Real, Invented, Not a cycle and All sum to 100. *Real*: every step of the cycle is an edge; *Invented*: a step is not an edge; *Not a cycle*: a length-2 or retraced walk. *Edge count*: the answers under All that argue from the number of edges, identified by their wording. The valid arguments are Real and Edge count, which is valid because 40 nodes with at least 40 edges must contain a cycle. Bold (Real, Invented only): differs from the same model without a primer (paired, BH $q < .05$).

sentence’s gap on plain 4B’s neighbor sets lies in $[-2.0, +2.5]$, inside the ± 3 -point margin. The degree primer’s gap on plain 1.7B’s neighbor sets is not significant ($q = .24$), but its interval reaches $+6.2$, so the two are not shown to be level.

Thinking mode (§4.5). Thinking models check stated values against their own count. Under the degree primer, thinking 1.7B still lists the neighbors in 90% of responses at $p \geq .35$; when their count conflicts with the stated value, thinking 1.7B and thinking 4B end on the stated value in 66% and 93% of cases. In the follow-ups, clustering gains thinking 1.7B 2.5 points at $p \leq .50$ ($q = .043$)

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.2	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.2	+1.2	-0.2	+1.0	+0.2	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.8	-2.5	-3.5
	1.7T	81.75	-0.8	+0.8	+5.0	-12.2	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.2	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.8
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 8: Complete main-sweep truncated shares: no-primer baseline (percent of all prompts) and paired changes (percentage points), for the same queries as Table 5. Each of its cells’ *wrong* share equals 100% minus its correct and truncated shares. P/T: plain/thinking; components and filler are diagnostics. No significance marks are assigned to truncation contrasts.

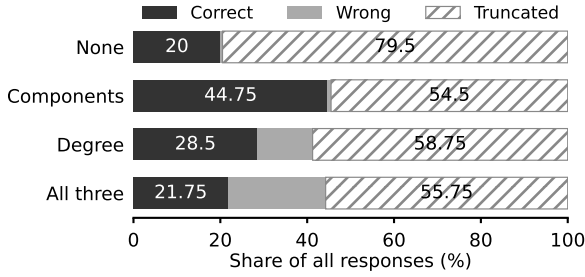


Figure 5: Thinking 4B on edge count: correct, wrong and truncated shares of *all* responses. Conditions from top: none, components, degree, all three.

and loses 5.4 at $p \geq .65$ ($q = .0015$), where filler loses 11.4. Of the answers a condition changes, 44.4–64.5% for thinking 1.7B and 83.3–95.2% for thinking 4B involve a response that ran out of tokens; Figure 5 shows this for thinking 4B’s edge count.

What responses rest on (§4.6). Clustering and RWSE change the joint share with $q \geq .16$ in every arm. Plain 4B’s is 85.75% with no primer, 76.50% with degree and 95.00% with the component sentence; the degree primer raises plain 1.7B’s from 54.00% to 62.50%. Table 7 classifies every correct, finished cycle-check answer by the argument it cites. For example, on one $p = .20$ graph plain 1.7B without a primer answers with $0 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 14 \rightarrow 13 \rightarrow 15 \rightarrow 16 \rightarrow 18 \rightarrow \dots \rightarrow 38 \rightarrow 39 \rightarrow 0$, which climbs the ids almost in order; steps such as $15 \rightarrow 16$, $21 \rightarrow 25$

and $38 \rightarrow 39$ are not edges. With thinking and no primer, valid arguments back 78.0% and 95.8% of all prompts. The degree primer raises plain 4B’s valid share from 58.5% to 76.5%. Truncation accounts for 21.2 of the 21.8 points it costs thinking 1.7B but only 8.2 of the 17.5 it costs plain 1.7B (Tables 5 and 8). RWSE, all three statistics and filler lower plain 4B’s valid share to 33.2%, 45.2% and 45.8% ($q < .05$).

Node count. The encoding’s first line lists the 40 node ids, so node count measures which number a model writes, not reading. Plain 1.7B answers 39, the last id, on 68.5% of finished responses without a primer and on 1.2% under clustering, hence Table 5’s large node-count effects.

C Pilot and the Clustering Screen

Pilot. On the published GraphQA graphs (5–19 nodes), Gemma 4 E4B/12B and Qwen3-8B/14B, each in plain and thinking mode, were near ceiling without a primer on most tasks; only edge count separated them. This is why the main study uses 40-node graphs and smaller models.

The clustering effect. To investigate the “clustering effect”—an unexpected performance gain observed during an initial 40-cell screening of Qwen3-1.7B on node degree tasks—we conducted a broader evaluation (Table e4tab:clust). We tested the effect on unseen graphs containing 20 to 160

Test	Gain	p or q
Screened graphs (700)	+1.29	$p = .47$
Unscreened graphs (2,100)	+2.1	$p = .031$
$p \leq .50$ (1,600 pairs)	+3.8	$q = .0035$
Fresh seed	+4.2	$p = .00055$ (.022 corr.)
Against filler	+3.2	$q = .014$
Fixed mean degree (3,200)	+6.2	$q < .001$
$p \geq .65$ (1,200 pairs)	-0.7	$q = .57$
Qwen3-8B (3,200)	-1.0	$q = .05$

Table 9: The clustering effect on node degree against no primer (against filler where the row says so), in points, for plain Qwen3-1.7B except the last row, which is Qwen3-8B. *Screened*: graphs the screen also used; *fixed mean degree*: the 20–160-node grid. *corr.*: Bonferroni over the screen’s cells.

nodes (at a fixed mean degree), utilizing fresh random seeds, and compared it against a filler text control. We also evaluated its generalization to high-density graphs and the larger Qwen3-8B model. The performance gain successfully transfers to unseen graphs and fresh seeds at low densities, but disappears on high-density graphs and does not occur in Qwen3-8B.

Further analysis reveals this improvement is an artifact of positional bias rather than actual graph clustering. The model’s responses never explicitly mention clustering, and the accuracy gain is independent of the queried node’s printed clustering coefficient ($p = 0.50$). Instead, the benefit stems from where the queried node appears in the prompt’s node list. Performance heavily favors nodes appearing early in the list: across all seven densities, we observe point gains of +5.01, +0.75, and -0.11 for nodes in the first, middle, and last thirds of the list, respectively. After adjusting for degree, clustering, and density, performance drops by 1.74 points for every 10 positions further down the list ($p = 0.018$). Neither filler nor component controls exhibit this positional gradient.

Finally, the clustering format alters how the model parses lists, resolving 224 previous neighbor-list copying errors while introducing 176 new ones. All formatting interventions reduce the number of missed neighbors compared to the unformatted baseline (which misses 1.49 neighbors per response). Missed neighbors drop to 0.86 with the clustering format, while the components and filler formats yield similar reductions (0.94 and 1.05, respectively, on the mid-density graphs, $p = .35$ –.50).